

Detecting Dialogue Hallucination using Graph Neural Networks

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Abstract—Even though large language models (LLMs) accumulate tremendous knowledge, dialogue systems built with LLMs induce hallucinations, leading to the generation of non-factual responses. How to provide proper references to achieve interpretable hallucination detection is a key issue that needs to be addressed. In this paper, we propose a graph neural network (GNN)-based method to achieve high-performance and interpretable hallucination detection for domain-specific dialogue systems. The method involves performing graph matching between a reference knowledge graph obtained from a knowledge database and a response knowledge graph extracted from the response to detect non-factual responses. By comparing with strong baselines, our method achieves a recall improvement of up to 11% and infers the cause of hallucinations with a probability of over 79%.

Index Terms—Dialogue system, hallucination detection, graph neural networks.

I. INTRODUCTION

LLMs have significantly improved the performance of dialogue systems [1]–[5]. In these studies, the LLM-based dialogue system is provided with some knowledge to generate an informative response text. However, such systems suffer from the problem of generating nonexistent or incorrect content. This phenomenon is referred to as a “hallucination” [6].

While there have been several works aimed at detecting hallucinations to improve knowledge-based text generation [7]–[10], the references required for detection are not well researched. Many existing works either use the knowledge that was given explicitly for generating text (gold knowledge) or manually provide a specific reference for hallucination detection [9]–[12]. Hence, existing works face difficulties in providing references. Regarding the use of gold knowledge as a reference, it may not always follow the knowledge contained in the response. This is because modern dialogue systems can inherit tremendous knowledge from LLMs and use it to generate a text [13], so knowledge other than Gold Knowledge is also included in the response text, resulting in the issue of insufficient referencing. Furthermore, since ChatGPT enabled to generate informative response text without explicitly provided knowledge [14], existing works cannot provide a reference. As a result, such a method is limited to accurately detect a narrow range of hallucinations.

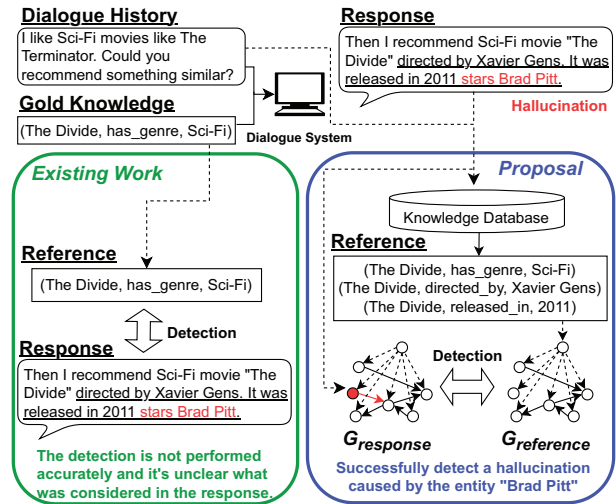


Fig. 1: Prior works generate references w/o response knowledge. In contrast, our proposed GNN-based method constructs two KGs with rich response knowledge, leading to high-performance and interpretable hallucination detection.

For example, regarding the example shown in Fig. 1, while the gold knowledge given to the dialogue system is only a knowledge triple (The Divide, has_genre, Sci-Fi), the response text contains more statements, such as “Xavier Gens directed The Divide”, “The Divide was released in 2011” and “Brad Pitt appeared in The Divide”. Here, the statement “Brad Pitt appeared in The Divide” is incorrect, while the other statements are correct. However, existing works cannot accurately consider these statements for detecting hallucinations. If we treat these statements as hallucinations just because the knowledge cannot be confirmed by the gold knowledge, it will limit the ability of LLMs to utilize knowledge. Additionally, there is another problem here, which is interpretability. Many methods use natural language to input the gold knowledge and the response text into LLMs and detect hallucinations in an end-to-end manner [7]–[10]. The problem here is the lack of clarity on what is considered in generating the response

text, which results in low interpretability of the causes of hallucinations. This makes it difficult to contribute to the improvement of dialogue systems.

One solution to the problems of providing references is to automatically collect the references needed to detect hallucinations based on the response text, but such methods have not yet been well-studied. Regarding the interpretability problem, analyzing the parameters of LLMs may be a potential solution, but such methods have not yet been established. To alleviate such problems, we focused on utilizing knowledge graphs (KGs), which have recently gained attention in the context of explainable AI [15], [16]. KGs are graphs composed of knowledge triples as structured representations, and they can be effectively utilized by graph neural networks (GNNs) [17], [18].

In this paper, we propose a GNN-based graph matching method to achieve high performance and interpretability in detecting hallucinations. The technical key of our proposed method is extracting knowledge from the response text and representing them in a knowledge graph. This clarifies the necessary reference and calculates useful differences between the response and the reference by GNN-based graph matching. The right side of Fig. 1 shows an overview of our method. Our method alleviates the problem with reference by automatically constructing two KGs: a response knowledge graph $G_{response}$ and a reference knowledge graph $G_{reference}$. $G_{reference}$ is composed of knowledge retrieved from a knowledge database based on $G_{response}$. Leveraging a knowledge database and knowledge graphs enables our method to achieve a wider and more accurate detection of hallucinations than using gold knowledge.

Furthermore, we also alleviate the interpretability problem. In our method, constructing $G_{response}$ provides a clearer understanding of what knowledge in the response text was considered for the detection. Additionally, our method can infer the cause of hallucinations by matching the two KGs. To perform the matching, we utilize the node features generated by a relational graph attention network (RGAT) [19] of the two graphs. The differences calculated using those node features contribute not only to interpretability but also to the improvement of detection performance.

To evaluate our method, we prepare a dataset of 800 responses annotated based on all the knowledge contained in a knowledge database as to whether response texts contain hallucinations. By comparing with strong baselines, our method achieves a recall improvement of up to **11%** by accurately detecting more hallucinations and infers the cause of hallucinations with a percentage of over **79%**. Our main contributions are: 1) We construct two types of KGs that include response knowledge to enrich the reference for detecting hallucinations; 2) We propose a GNN-based matching method to achieve interpretable detection.

II. RELATED WORK

Dialogue systems. The performance of dialogue systems has been improved by utilizing LLMs. Modern neural dialogue

models, whether task-oriented or open-domain, can generate more informative response texts [1], [2], [14]. KGs have also gained prominence in the context of dialogue models that provide informative responses [3]–[5]. Reference [5] proposed a method that allows a single transformer model to walk directly on a large scale KG and generate response texts. In addition, publicly available KGs have been proposed [20], [21] and more dialogue models will be based on tremendous knowledge of the KGs.

Detecting hallucinations. According to previous works [6], [10], [22], there are mainly two types of hallucinations: intrinsic and extrinsic hallucinations. Intrinsic hallucinations are defined as the ones that can be appropriately detected based on the gold knowledge K_n . Extrinsic hallucinations are defined as the ones whose faithfulness cannot be determined based on K_n alone. Many existing studies on detecting hallucinations only address intrinsic hallucinations [7], [8]. Our proposed method can accurately detect a wider range of hallucinations than related work because it can handle both intrinsic hallucinations and some extrinsic hallucinations using a knowledge database. To the best of our knowledge, this study is the first to address hallucination detection within the scope of a specific knowledge database. Detecting hallucinations according to a specific dataset is important in the development of real-world conversational agents. For instance, it enables the creation of reliable chatbots by ensuring they provide knowledgeable and accurate utterances about the company's services while avoiding the provision of uncertain knowledge that is not guaranteed by the prepared dataset.

There are methods [23], [24] to verify the factual correctness of generated texts by extracting knowledge from documents generated by dialogue models. However, it differs from our method in that it does not utilize knowledge databases or take advantage of knowledge triples using GNNs. The work most relevant to our method is that by [10]. The authors of this study address the detection of hallucinations in dialogue. Our proposal differs from them in two ways: first, our method can accurately detect a broader scope, and second, our method is able to partially capture the nature of hallucinations and provide some interpretability using GNNs.

III. PROBLEM FORMULATION

In this paper, as a knowledge-grounded dialogue, we consider data with a dialogue history $H = (x_1, \dots, x_{n-1})$ consisting of response texts x and a set of gold knowledge triples $K_n = (t_1, \dots, t_m)$ for generating the next response text x_n following the dialogue history. Here, t represents a triple format $\langle s, p, o \rangle$, where $s, o \in \mathcal{V}$ and $p \in \mathcal{R}$. In this work, we focus on the use of multi-relational KGs $G = (\mathcal{V}, \mathcal{E}, \mathcal{R})$, where $\mathcal{V}$ is a set of nodes, $\mathcal{E}$ is a set of edges, and $\mathcal{R}$ is a set of relation types. G consists of a set of knowledge t . We train our model to predict binary hallucination labels y for each response according to the following equation: $\hat{y} = f_\theta(G_{response}, G_{reference})$.

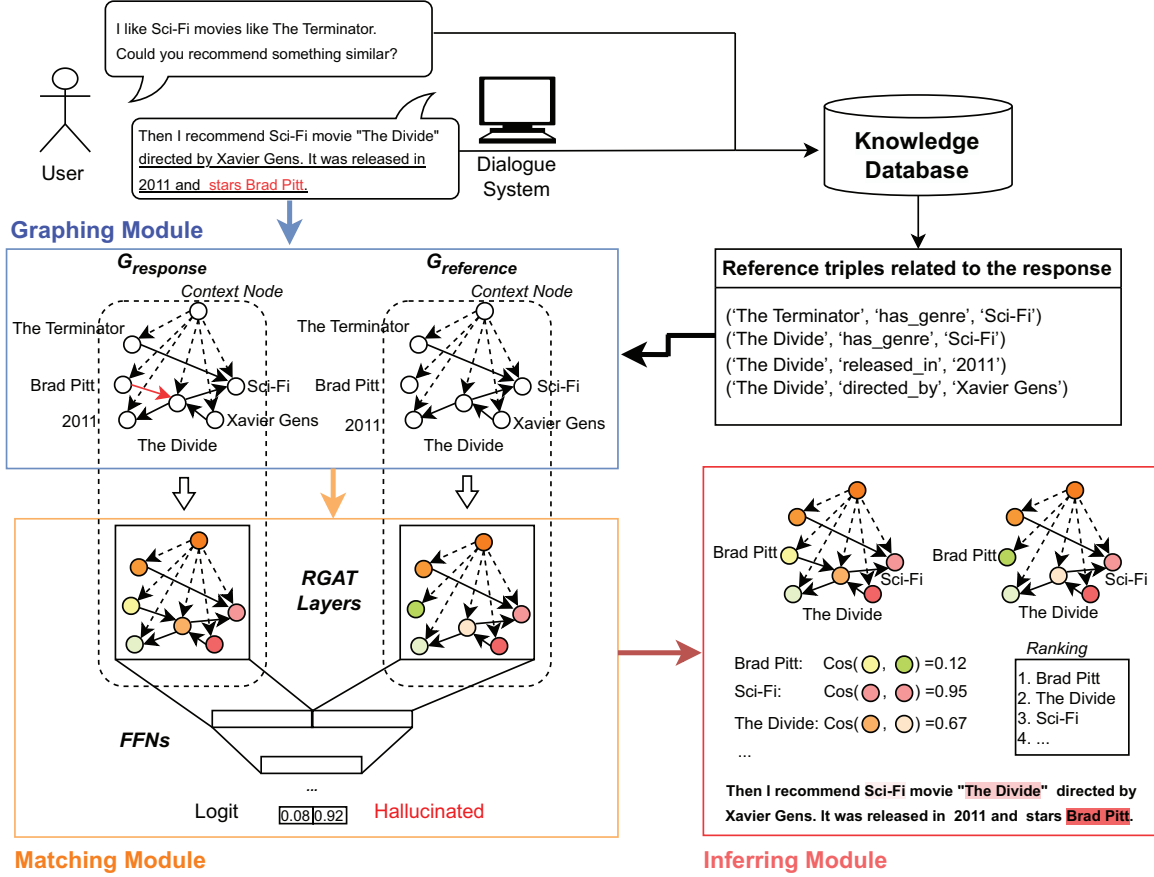


Fig. 2: Given the dialogue history and the response, two KGs are constructed ($G_{response}$ consisting of knowledge extracted from responses, and $G_{reference}$ consisting of the knowledge retrieved from a database). By matching these two graphs in the matching module and the inferring module, the response text is binary classified, and causal entities are inferred.

IV. METHOD

We present the details of the proposed method in Fig. 2. We designed a hallucination detection model using a knowledge database. For this purpose, we proposed a model consisting of a module for constructing two KGs and two modules for matching and inference based on those graphs. The graphing module automatically constructs two graphs, namely $G_{response}$ and $G_{reference}$. The matching module and the inferring module calculate the difference between these graphs using a relational graph attention network (RGAT) [19], which contributes to the binary classification and inference of the cause of hallucinations.

A. Graphing Module

This module uses the response text and the dialogue history as input and extracts the knowledge stated by the response text to generate the two graphs that are input into the matching module. For that purpose, this module first performs named entity recognition (NER) on the dialogue history and the

response text. We used spaCy¹ and string matching for NER. Here, for the extracted entities, let e^H be an entity in the dialog history and e^{x_n} be an entity in the response text. Then, relationship extractions (REs) are performed on each entity pair $(e_l^H, e_m^{x_n})$ or $(e_l^{x_n}, e_m^H)$ for predicting the relational type $\hat{o}$, where $e_k^{x_n} \in e^{x_n}$ and $e_k^H \in e^H$. RE is not performed between the same entities. The relational type o represent relationships between entities, such as "has_genre". For the prediction of o , we utilized the RE model created following the approach of [25], [26].

Note that based on the knowledge database, if the predicted relation type is not related to the head entity or tail entity, it will be excluded. This indicates the case where knowledge $t = \langle s_l, p, \hat{o} \rangle$, which includes a specific head entity s_l , some tail entity p , and a predicted relational type $\hat{o}$, is not present in the knowledge database. This process is designed to mitigate RE errors. The KG consisting of the knowledge extracted by such processes is denoted as $G_{response}$.

Next, we construct a graph $G_{reference}$ with the same

¹<https://spacy.io/>

nodes (entities) as those in $G_{response}$. For $G_{reference}$, all possible edges of connection between these entities are extracted and set from the knowledge database. Finally, a context node is added to each graph and is connected to all nodes. This context node is used not only to consider contextual information but also to avoid relying solely on the quality of the knowledge triple extraction.

B. Matching Module

The matching module uses the two graphs created in the graphing module to identify whether the response text contains a hallucination. For this purpose, the features of each node are first initialized by RoBERTa encoder [27], which encodes the concatenated text of the dialogue history H , response text x_n , and knowledge contained in $G_{reference}$. Specifically, the contextual information nodes are initialized using the embedding of the CLS token, and other nodes are initialized by the embedding of the corresponding entity tokens.

After that, each graph node feature is updated using the RGAT, and the graph features $\mathbf{h}$ are generated by a global average pooling layer. Note that each edge is processed as an undirected edge. Node embeddings in the l -th layer, attention weights α in the RGAT layer, and a graph feature $\mathbf{h}$ are computed as

$$\{\mathbf{v}_0^l, \dots, \mathbf{v}_{|V|}^l\} = RGAT(\mathbf{v}_0^{l-1}, \dots, \mathbf{v}_{|V|}^{l-1}), \quad (1)$$

$$\alpha_{i,j}^{(r)} = \frac{\exp(\mathbf{a}_{i,j}^{(r)})}{\sum_{r' \in \mathcal{R}} \sum_{k \in \mathcal{N}_{r'}(i)} \exp(\mathbf{a}_{i,k}^{(r')})}, \quad (2)$$

$$\mathbf{h} = \frac{1}{|V|} \sum_{n=1}^{|V|} \mathbf{v}_n^L, \quad (3)$$

where r represents a relation type, $\mathcal{R}$ denotes a set of relations, $\mathbf{v}_i$ represents a feature vector of node v_i , $\alpha_{i,j}^{(r)}$ represents an attention weights for an edge with respect to the relation r between v_i and v_j in the RGAT layer, L represents the number of layers in the GNNs, and $\mathbf{h}$ is the average pooling of the last layer. The combined vector of each graph feature $\mathbf{h}$ is input to Feed Forward Networks (FFNs) consisting of fully connected layers to predict the label y through binary classification $\hat{y} = FFNs(\text{concat}[\mathbf{h}_{response}, \mathbf{h}_{reference}])$.

C. Inferring Module

Identifying an entity that causes a hallucination is a complex task. Here we approach this task assuming that a hallucination affects multiple entities in a text. For example, in Fig. 2, “Brad Pitt” did not appear in “The Divide”. In this case, so both “Brad Pitt” and “The Divide” could be the cause of hallucinations. Hallucinations arise as a result of multiple entities influencing each other. For the reasons mentioned above, we do not adopt an approach that would identify only a specific entity as the cause. Instead, we adopt an approach that captures the degree of hallucinations. Our proposed method captures the nature of hallucinations by computing graph features using RGAT with the same parameters for

$G_{response}$ and $G_{reference}$. Here we focus on the differences between the respective node representations. For each entity, our method computes the cosine similarity between the two node representations computed by each graph and predicts that the smaller it is, the more likely it is to be the cause of hallucinations. For example, in Fig. 2, “The Divide” has several edges that exist in both $G_{reference}$ and $G_{response}$. On the other hand, “Brad Pitt” has only one edge present in $G_{response}$, excluding the contextual node. Due to such differences in edges between the two graphs, the features related to the entity “Brad Pitt” have the lowest cosine similarity. Then, we assume that this similarity can be treated as a measure of the degree of hallucinations.

V. EXPERIMENTS

A. Experimental Design

There are no benchmarks for the tasks we focus on. Therefore, we designed experiments and created data to answer the following research questions:

- **RQ1: Does the proposed method accurately detect a wider range of hallucinations?**
- **RQ2: Is the method of comparing the two knowledge graphs effective?**
- **RQ3: Can the proposed method explain the causes of hallucinations?**

For RQ1, we compared the detection performance of our method with that of methods using gold knowledge K_n . We prepared a baseline model based on RoBERTa [7]–[10]. In addition, we also prepared a baseline model that uses knowledge retrieved from the knowledge database as a reference in the same way as our approach. This allows us to answer RQ2 by confirming the effectiveness of our comparing module. We answered RQ3 by using our inferring module to infer the entities that the annotators labeled as the cause of hallucinations. For these experiments, we prepared four dialogue models and annotated whether their generated response texts contain hallucinations through crowdsourcing.

B. Knowledge-Grounded Dialogue Dataset

All experiments were conducted using OpenDialKG [28]. OpenDialKG is currently the only publicly available dataset that contains knowledge-grounded human-to-human dialogues annotated with knowledge triples used in the response text. The topics in this dataset are movies, books, sports, and music. They used a subset of Freebase Easy data [20] as its knowledge database, which consists of approximately 1.19M pieces of knowledge. No official split is provided in the original dataset, and thus, we randomly split the dataset into 80/10/10 for training/validation/testing, respectively. Since there are many chit-chat data, only data with annotated knowledge triples were used. As a result, the numbers of data points for training, validation, and testing are 62130, 7659, and 7641, respectively. In OpenDialKG, there are 337 types of relations, and the trained RE model achieved a prediction accuracy of 90.3% for relational types on the test set.

TABLE I: Results of the human annotation for each dialogue model response texts (200*4). “Faithful” indicates that the response text doesn’t hallucinate. “Info.” indicates the non-chit-chat response texts.

Model	Faithful	Info.
GODEL	143 (18)	172
T5	150 (21)	178
RoBERTa	65 (19)	127
ChatGPT	157 (53)	195

In addition, to train models, we created pseudo-hallucination data from knowledge-grounded human dialogues and evaluated the model’s performance on crowdsourced annotated data. Preparing a sufficient amount of human-annotated data for training is costly. Therefore, we processed OpenDialKG dataset, replacing entities to create a pretraining dataset [10]. Note that replacements were randomly selected from entities of the same type (person name, work name, year, etc.) recognized using spaCy. Each text had one or two replacements, and we made a pretraining dataset with 30% hallucination data.

C. Dialogue Models

To evaluate the ability of the proposed model to detect hallucinations in practice, we generated response texts with four dialogue models. We used both traditional and state-of-the-art methods: RoBERTa Encoder-Decoder, T5 [29], GODEL [1], and ChatGPT [14]. The conditional probability for response text ($\mathbf{x}_n = \{w_1^n, \dots, w_K^n\}$) generation can be expressed as follows: $p(\mathbf{x}_n | \mathbf{H}, \mathbf{K}_n) = \prod_{k=1}^K p(w_k^n | w_1^n, \dots, w_{k-1}^n, \mathbf{H}, \mathbf{K}_n)$. We trained the response generation task for three models other than ChatGPT, following the equation mentioned above. For ChatGPT, we provided a prompt instructing it to generate a response based on $\mathbf{H}$ and $\mathbf{K}_n$. We used the cross-entropy loss function and trained them through teacher-forcing.

D. Manual Annotations

We use Prolific² to annotate the response texts generated by the four dialogue models. In our task, we need to evaluate whether a response text is hallucinated using all the knowledge contained in the knowledge database. However, such work is costly, so the data were annotated in two steps. In the first step, 800 response texts generated by the four dialogue models were evaluated through crowdsourcing. Annotators were provided with the dialogue history $\mathbf{H}$ and gold knowledge $\mathbf{K}_n$ to evaluate the faithfulness of response text $\mathbf{x}_n$. If the response text contained a hallucination, the annotators were asked to answer which entity caused the hallucination. In the second step, the data labeled as hallucinations in the first step were further evaluated. We re-evaluated these data using all available knowledge related to all entities in $\mathbf{H}$ and $\mathbf{x}_n$.

Each response text was evaluated by at least five annotators. Note that some data for filtering annotators were inserted to

ensure the quality of the evaluations. The evaluation results are shown in Table I. “Faithful” indicates that no knowledge in the response text is hallucinated, and “Info.” indicates that the response text contains at least one piece of knowledge, i.e., it is not a chit-chat response such as greetings. We excluded non-informative data points, so a total of 672 data points were used for the experimental results in Section VI. The number of data points that were labeled as a hallucination in the first step but were not in the second step is 111. In other words, at least for these 111 data points, detection based solely on the gold knowledge $\mathbf{K}_n$ cannot be appropriately handled. The number of data for each model is indicated in parentheses in Table I. In particular, ChatGPT has a strong tendency to mention knowledge other than $\mathbf{K}_n$ given in the prompt, and there were 96 such data, of which 53 were “Faithful” data. We created five folds using the annotated data and performed cross-validation.

VI. RESULTS

TABLE II: Cross-validation results on baseline and proposed method using human-annotated data. “Ours” and “Baseline” used reference knowledge automatically retrieved from a knowledge database (Section IV-A), and Baseline($\mathbf{K}_n$) used gold knowledge. Our method generally demonstrates superior performance compared to the baselines.

Pretraining	Method	Acc.(%)	Prec.(%)	Recall(%)
✓	Ours	85.3±3.6	74.0±6.3	63.2±6.8
	Baseline	78.4±3.3	67.2±5.9	53.3±8.6
	Baseline ($\mathbf{K}_n$)	79.8±2.5	72.2±2.1	51.6±5.5
-	Ours	81.3±4.2	61.2±6.9	70.6±8.2
	Baseline	75.9±2.9	62.0±5.0	50.9±11.8
	Baseline ($\mathbf{K}_n$)	75.7±0.9	71.7±8.0	40.7±11.2
	Human ($\mathbf{K}_n$)	79.0	58.4	84.6

TABLE III: Comparison results of hallucination detection for each dialog models. All values are expressed in percentage (%).

Model	Method	Acc.	Prec.	Recall	F1
GODEL	Ours	92.4	81.4	71.9	75.2
	Baseline ($\mathbf{K}_n$)	84.7	54.2	36.0	42.6
T5	Ours	86.6	56.6	54.1	52.7
	Baseline ($\mathbf{K}_n$)	83.2	69.3	20.5	30.3
RoBERTa	Ours	82.4	89.3	74.0	79.8
	Baseline ($\mathbf{K}_n$)	81.9	83.5	84.5	83.2
ChatGPT	Ours	83.2	63.5	46.6	54.8
	Baseline ($\mathbf{K}_n$)	71.5	58.7	49.9	53.4

Table II shows the main results along with the standard deviation across five folds in cross-validation. We prepared a baseline model with LLMs, following related studies that addressed similar tasks [9], [10], [30]. We used RoBERTa and fine-tuned it. A dialogue history, a response text and a

²<https://www.prolific.co/>

reference are used as input in this model to predict the hallucination label $\hat{y} = f_{\theta}(\mathbf{H}, \mathbf{x}_n, \text{Reference})$. For calculating scores, data that included a hallucination were labeled positive, and data that did not were labeled negative. “Ours” and “Baseline” used reference knowledge automatically retrieved from a knowledge database (Section IV-A), and “Baseline($\mathbf{K}_n$)” used gold knowledge that were input into the dialogue model to generate responses. “Human” represents the average score of the annotators who participated in the first step (Section V-D). Therefore, these annotators only refer to gold knowledge $\mathbf{K}_n$, resulting in low precision.

RQ1: Does the proposed method accurately detect a wider range of hallucinations?

Overall, the performance of all models improved through pretraining, demonstrating that the automatically generated pretraining dataset is effective. In the results with pretraining, our model showed higher accuracy and precision than those of the baseline and human scores, indicating the reliability of our model’s detection results. One of the reasons for our method’s high accuracy is its ability to utilize the automatically constructed reference to correctly identify “True Negatives”. Our method, which automatically constructs references based on response texts, can detect a wider range of hallucinations and has a recall score of over 11% higher than the baseline using $\mathbf{K}_n$. Furthermore, the results in Table III demonstrate that our method can detect hallucinations in dialogue models with consistently higher accuracy than baselines. Note that, as discussed in RQ2, it is important to compare the knowledge indicated by the response texts with the reference knowledge. Also, one of the reasons for Ours having a lower F1 score compared to Baseline($\mathbf{K}_n$) is that the RoBERTa-based dialog model generates more grammatical errors than other dialog models. Our approach may fail to extract triples contained in grammatically unnatural sentences, resulting in the inability to detect some hallucinations, and consequently leading to lower Recall compared to Baseline($\mathbf{K}_n$).

RQ2: Is the method of comparing the two knowledge graphs effective?

When comparing the baseline using the knowledge database with the baseline using $\mathbf{K}_n$, their performance is similar. However, the proposed method, which extracts response knowledge and performs matching with references using the GNN, demonstrated higher performance compared to the baseline. This leads to the conclusion that hallucination detection requires more than just providing reference knowledge and an effective matching mechanism is necessary.

RQ3: Can the proposed method explain the causes of hallucinations?

We evaluated the inference ability of our method using entities labeled by annotators as the causes of hallucinations. We computed the ranking of the entities predicted by our hallucination entity inference method (Section IV-C), using a mean reciprocal rank-based metric (MRR). The MRR-based

TABLE IV: Results of the evaluation of the inferring performance of entities causing a hallucination.

Method	Dialogue Model	MRR	Top@1
Ours	GODEL	0.904	0.766
	T5	0.867	0.740
	RoBERTa	0.966	0.917
	ChatGPT	0.799	0.770
Unsorted	GODEL	0.795	0.616
	T5	0.638	0.320
	RoBERTa	0.888	0.812
	ChatGPT	0.547	0.486

metric evaluated the rank of causal entities in the predicted ranking by the model. The MRR score is calculated based on the following formula:

$$\text{MRR} = \frac{1}{|\mathbf{e}|} \sum_{n=1}^{|\mathbf{e}|} \frac{1}{\text{Rank}(e_n)}, \quad (4)$$

where $\text{Rank}(e_n)$ represents the ranking of entity e_n in the predicted ranking, and $\mathbf{e}$ is the set of entities labeled by the annotator. The gold ranking was created by sorting the entities with more votes by annotators to have a higher rank. The MRR scores were normalized by the maximum score per dialogue model data. The ranking was assessed by comparing it to a list of entities (Unsorted) arranged according to their appearance in the text. The results are shown in Table IV, where our method is effective in inferring the causal entities. According to the Top@1 score, our method can correctly identify the cause of hallucinations with an average accuracy of 79.8%. The average number of response entities extracted by NER is 2.7, and the average number of hallucination entities labeled by annotators is 1.5. In particular, ChatGPT’s average number of response entities is 4.5, which is higher than that of the other models, and our method has shown the most improvement in scoring when ranking entities. From the above results, our method is capable of generating interpretable node representations during the process of hallucination detection.

VII. CONCLUSION & LIMITATION

We proposed a GNN-based graph matching method to detect hallucinations in dialogue systems. Our method constructed the retrieved reference KG and the extracted response KG. By utilizing GNN to match those two KGs, our method achieved high-performance and interpretable hallucination detections. We compared our method with strong baselines on a crowdsourcing dataset, and the results demonstrated that our method detected much more hallucinations accurately than baselines. Furthermore, our method inferred the causes of hallucinations by tracking the learned features of our model, suggesting its efficacy of interpretability.

For one limitation, as we mainly dealt with entities mentioned in the response text, our method cannot handle the knowledge that is implied without being explicitly stated (e.g.

“The director of these two films is the same.”). If we can incorporate a method to extract implicit knowledge triples into the graphing module, our approach would be able to detect even more hallucinations. Another limitation of our approach is its high dependence on the quality of the knowledge database. While this enables faithful hallucination detection based on evidence, it also requires manual maintenance. Developing reliable chatbots necessitates a trustworthy knowledge database, making the construction of high-quality knowledge databases an important research topic. Furthermore, there is room for improvement in relation extraction accuracy. Although the RE model achieved high accuracy in predicting relational types, this was partly due to the occurrence frequency bias towards specific relational types in OpenDialogKG. Specifically, the nine most frequent relation types accounted for over 75% of the total occurrences. Improving the prediction accuracy for the less frequent ones remains a future challenge. We are planning to address these issues in our future work.

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